

InkProf - When colours have to be right

Why InkProf is needed

For photographers, a print should carry the expression of the image. For printing businesses, it must also deliver consistent quality at a manageable cost. Poor calibration and incorrect settings cost time, paper and ink. A calibrated and verified colour workflow can reduce reprints and waste.

InkProf brings measurement, profiling and print verification together. Open source reveals the methods; saved measurements, settings and decisions make results traceable. This helps you find errors and explain results to clients. Transparency enables scrutiny; it does not guarantee correct colour.

Documented comparisons with standards-based limits support quality assessment without certifying full compliance. Verification results help establish what the chosen combination can achieve: a profile cannot reproduce colours, black density or contrast beyond the capabilities of the printer, paper and ink.

From a saved profile to a documented decision

InkProf helps you understand how a profile was created, how it was checked and why it is approved or needs further improvement.

What is InkProf?

InkProf is a project-based MATLAB app for profiling RGB-driven printers. It creates colour targets, saves TIFF16 files, handles spectral measurements, builds ICC profiles and compares profile predictions with measured prints.

Who is it for?

For photographers and printing businesses that want reliable colour, less waste and a documented basis for client approval. Each printer, paper, ink and settings combination is managed as its own project.

TIFF16 for printing • Measurement in the app • ICC with PDF and HTML reports

Open source and technical foundation

InkProf uses base MATLAB, with no mandatory add-on toolboxes, together with Python and ArgyllCMS. JSON is used internally, with T11, T12 and T13 as core exchange formats. InkProf is open source under GNU GPL version 3 or later (GPL-3.0-or-later). You may inspect, modify and share the code under the licence terms. External dependencies have their own terms.

THE WORKFLOW

From target to profile

The app guides you step by step. A later operation is enabled only when its required inputs are complete and still valid.

- 01 Create or open a project**
Each project has its own folder. Create or import the RGB target and select a layout for the measurement sheets.
- 02 Save TIFF16 and print separately**
The app saves 16-bit TIFF files, the layout and matching measurement inputs. You print separately and allow the sheets to dry. The project can be closed while you wait.
- 03 Measure and review**
Measurements use a high-quality spectrophotometer, such as the i1Pro 2, to measure the spectral reflectance of the patches. Start measurement in the app or select a saved revision. Review and accept the result before locking the profiling inputs.
- 04 Build and check the ICC candidate**
The profile is built using ArgylCMS. Automatic iterations can compare candidates when development data are available. Numerical checks examine the model before the next step.
- 05 Check the actual print**
Save a verification target, print it and measure it in the app. The analysis compares the intended colour, the profile prediction and the measured colour using the Delta E 00 colour-difference metric.
- 06 Approve or improve**
Approve the profile for its intended use and save the ICC file with a final report. If needed, additional colour samples are created, measured and carried forward into the next iteration.

Printing and drying remain part of your practical work.

InkProf saves the files and project state. When the sheets are ready, open the existing project and continue with measurement.

This describes the implemented development version. Basic measurement has been tried with the i1Pro 2; the complete workflow needs verification with the actual instrument, paper and print settings.

FEATURES AND DELIVERABLES

A process you can trace

Projects that remember

The workflow state is saved in JSON, including the current step, iteration, dependencies and results. You can close the program and continue later. Missing or changed inputs block dependent operations. Repeating a step makes downstream results out of date until they are updated.

Traceable iterations and an accessible log

Each iteration is linked to its predecessor. The results log records events, parameters, files and errors. Checksums bind results to the correct inputs. Earlier results are retained, and a new candidate requires fresh checks and a new approval.

Measurement-based refinement and a 3D view

The analysis distinguishes measured colour error from disagreement with the profile model. It can propose additional RGB samples in areas that need investigation. A rotatable CIELAB view shows the profile-predicted verification colours; a still image can be included in the final report. The view shows selected verification points, not the printer's entire gamut.

ICC profiles and reports to save and share

You choose where to save the ICC profile and reports. The project retains its own copies. The PDF and HTML final report summarises the profile, measurement conditions, verification results, decisions and iteration history. Headers and footers identify the report, date, user and page count.

What the workflow delivers

Saved TIFF16 files for separate printing. An ICC profile approved for its intended use. PDF and HTML reports, plus a project that preserves the inputs, raw data and history.

Documentation with ISO as a reference

InkProf documents colour verification with reference to ISO 12647-2:2013 (offset) and ISO 12647-7:2016 (contract proofs). The tolerance example comes from MediaStandard Print 2018, published by Bundesverband Druck und Medien (bvdM), the German printing and media industry association. This publication summarises ISO requirements; it is not the ISO standard itself.

The documentation explains CIEDE2000 (Delta E 00), references, measurement conditions and reporting of the mean, median, P95 and maximum. It supports defining and assessing quality requirements with explicit sources. InkProf's own RGB targets and internal thresholds do not by themselves establish ISO conformity or certification; a formal claim requires the full testing specified by the applicable standard.

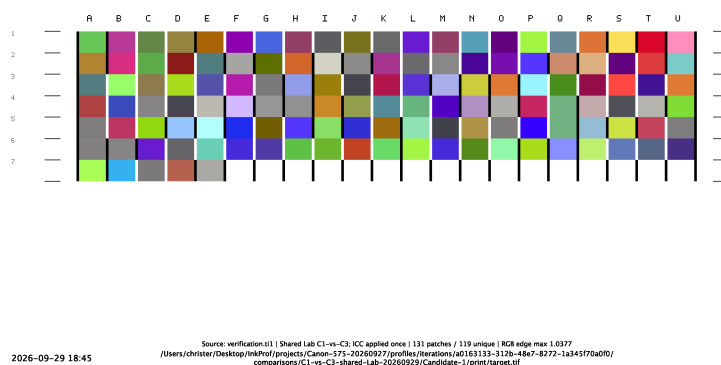
Sources: InkProf project documentation and the profiling workflow MATLAB guide, particularly pages 9-11 on ISO and sources. Project status: 2026-09-30.

EXAMPLES FROM INKPROF

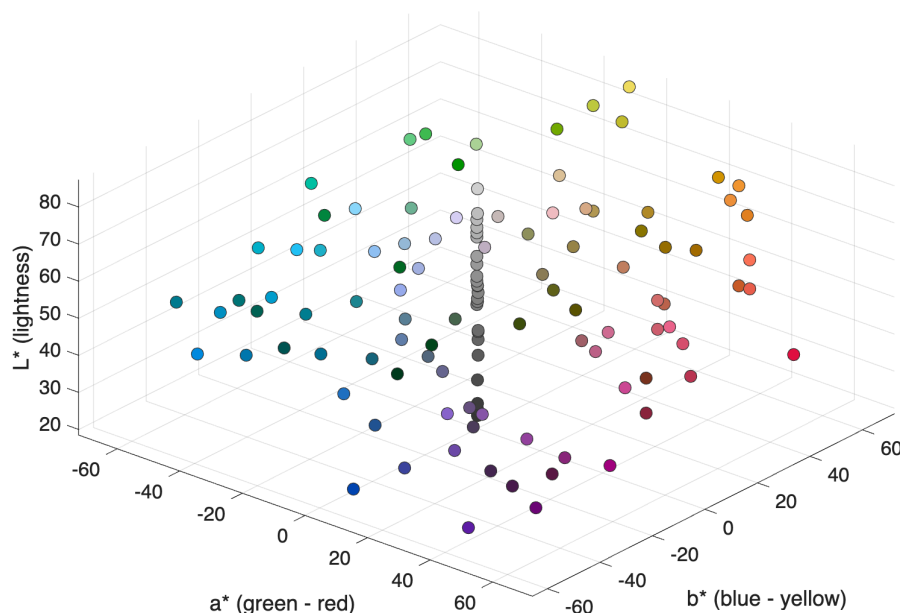
Colour target and 3D profile

These examples belong to the same verification target and ICC candidate for an Epson Stylus Pro 3880 on glossy paper.

InkProf Quality profiling of a RGB ink printer.



Patch target. A verification sheet with 131 patches, including repeats, contrast markers and position labels. This is a reduced preview of the saved TIFF16 file. Do not use the image in this presentation as a measurement print.



The profile's verification colours in 3D. The points show the ICC candidate's predicted colours in CIELAB D50: lightness L^* and the a^* and b^* colour axes. The view can be rotated in the app. It shows verification points, not measured values or the printer's entire gamut. Colours are display-adapted previews.

Image source: InkProf, Candidate-1 shared Lab comparison, 2026-09-29.